Intake of whole grain products and risk of breast cancer by hormone receptor status and histology among postmenopausal women.

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No clear relationship between whole grain products and risk of breast cancer has been established. In a large prospective cohort study, we investigated the association between intake of whole grain products and risk of breast cancer by tumour receptor status [oestrogen receptor (ER) and progesterone receptor (PR)] and tumour histology (ductal/lobular). It was further investigated whether the association differed by use of hormone replacement therapy (HRT). The study included 25,278 postmenopausal women participating in the Danish Diet, Cancer and Health cohort study (1993-1997). During a mean follow-up time of 9.6 years, 978 breast cancer cases were diagnosed. Associations between intake of whole grain products and the breast cancer rate were analysed using Cox's regression model. A higher intake of whole grain products was not associated with a lower risk of breast cancer. Per an increment in intake of total whole grain products of 50 g per day the adjusted incidence rate ratio (95% confidence interval) was 1.01 (0.96-1.07). Intake of rye bread, oatmeal and whole grain bread was not associated with breast cancer risk. No association was observed between the intake of total or specific whole grain products and the risk of developing ER+, ER-, PR+, PR-, combined ER/PR status, ductal or lobular breast cancer. Furthermore, there was no interaction between intake of whole grain products and use of HRT on risk of breast cancer. In conclusion, intake of whole grain products was not associated with risk of breast cancer in a cohort of Danish postmenopausal women. Copyright (c) 2008 Wiley-Liss, Inc.